

B.Sc. IV – Practical Mid Term Examination**Paper Code: STAMN-42**

Time: 2 Hours

Full Marks: 20

Instruction: Attempt any one of the following questions. Each carries equal marks.

Q1. Let $X \sim \mathcal{N}(\mu = 50, \sigma^2 = 25)$. A random sample of size $n = 25$ is drawn from this distribution, and the experiment is repeated independently 30 times.

Define the estimators:

$$T_1 = \bar{X}, \quad T_2 = \frac{X_1 + X_2 + X_3}{3}, \quad T_3 = \frac{X_1 + X_2 + \cdots + X_{10}}{10}$$

- (a) Compute all estimators.
 - (b) Show that all estimators are unbiased.
 - (c) Compute the variance of each estimator.
 - (d) Among all given estimators, which estimator is efficient estimator.
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Q2. A sample of size $n = 25$ is drawn from a normal population to study daily electricity consumption (in kWh). The sample mean is $\bar{x} = 18.5$ and the sample standard deviation is $s = 3.2$.

- (a) Construct confidence intervals for the population mean at 70%, 80%, 90%, and 95% levels.
 - (b) Interpret each confidence interval in context.
 - (c) Compute the length of each confidence interval.
 - (d) Plot the relationship between confidence level and interval length.
 - (e) Comment on how the interval length changes with confidence level.
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